6. A thin converging lens of refractive index 1.5 has a power of + 5.0 *D*. When this lens is immersed in a liquid of refractive index n, it acts as a divergent lens of focal length 100 cm. What must be the value of *n*?

[Ans: $n = 5/3$]

7. If the distance *D* between an object and screen is greater than 4 times the focal length *f* of a convex lens, then there are two positions for which the lens forms an enlarged image and a diminished image respectively. This method is called conjugate foci method. If *d* is the distance between the two positions of the lens, obtain the equation for focal length of the convex lens.

$$\left[\text{Ans: } f = \frac{D^2 - d^2}{4D}\right]$$

8. Prove that a concave lens can only form a virtual, erect and diminished image.

9. A point object is placed at 20 cm from a thin plano-convex lens of focal length 15 cm whose plane surface is silvered. Locate the position and nature of the final image.

[Ans: $v = -12$ cm]

10. Find the ratio of the intensities of lights with wavelengths 500 nm and 300 nm which undergo Rayleigh scattering.

[Ans: 81:625]

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